HW1

1. $f = 60$

$V_{rms} = 120\angle 0$

$v(t) = 120\sqrt{2}\sin(120\pi t)$

10Ω

i a. $Z = 10\Omega$; $10\angle 0\,\Omega$

b. $I = 12\angle 0\,A$ (RMS)

ii. 20mH

a. $Z = j(120\pi)(.02) = j2.4\pi\,\Omega$; $2.4\pi\angle 90^\circ\,\Omega$

b. $I = \frac{50}{\pi}\angle -90^\circ\,A$ (RMS)

iii 20mH and 10Ω

a. $Z = j2.4\pi + 10$; $12.524\angle 37.016^\circ\,\Omega$

b. $I = 9.582\angle -37.016^\circ$ (RMS)

iv 5mF

a. $Z = \frac{1}{j\omega C} = -j\left(\frac{1}{120\pi(5\times10^{-3})}\right) = -j530.516\,\Omega$; $530.516\angle -90^\circ\,\Omega$

b. $I = .226\angle 90^\circ\,A$ (RMS)

v. 5mF and 15Ω

a. $(15 - j530.516)\,\Omega = Z$; $530.728\angle -88.38^\circ$

b. $I = .226\angle 88.38^\circ\,A$ (RMS)

2. a. capacitor

b. inductor, avg power is positive. Power = V·I

c. resister for b, negative power is minimized, and c, Not all power is perfectly consumed